

API Reference — Supply Chain Disruption Dashboard

Base URL: `http://localhost:8000` **CORS:** Enabled for all origins **Map Center:** lat 38.8816, lon -77.0910 **Default Zoom:** 13

Base Layer Endpoints (call on app startup)

GET /arlington/bbox

Map configuration.

```
json
{
  "bbox": {
    "south": 38.827,
    "west": -77.172,
    "north": 38.934,
    "east": -77.032
  },
  "center": { "lat": 38.8816, "lon": -77.0910 }
}
```

GET /arlington/boundary

County outline. Draw as dashed cyan polyline.

```
json
```

```
{
  "name": "Arlington County",
  "outer_rings": [
    [
      { "lat": 38.827, "lon": -77.032 },
      { "lat": 38.828, "lon": -77.033 }
    ]
  ],
  "ring_count": 4
}
```

GET /arlington/roads/major

Motorways and trunks (I-66, I-395, US-50). Draw as orange/yellow polylines.

```
json
{
  "total_roads": 1614,
  "roads": [
    {
      "osm_id": 12345,
      "name": "Interstate 66",
      "highway_type": "motorway",
      "ref": "I-66",
      "coordinates": [
        { "lat": 38.885, "lon": -77.096 },
        { "lat": 38.884, "lon": -77.100 }
      ]
    }
  ]
}
```

GET /arlington/roads/secondary

Primary and secondary roads. Draw as green polylines, thinner.

Same format as /roads/major.

GET /arlington/railways

Freight rail lines. Draw as purple dashed polylines.

json

```
{
  "total_railways": 48,
  "railways": [
    {
      "osm_id": 67890,
      "name": "CSX Transportation",
      "railway_type": "rail",
      "usage": "main",
      "operator": "CSX",
      "coordinates": [
        { "lat": 38.85, "lon": -77.05 },
        { "lat": 38.86, "lon": -77.06 }
      ]
    }
  ]
}
```

GET /arlington/pois/gas-stations

Gas station markers. Draw as orange fuel pump icons.

json

```
{
  "total": 86,
  "stations": [
    {
      "osm_id": 11111,
      "name": "Shell",
      "brand": "Shell",
      "address": "1234 Columbia Pike, Arlington, VA",
      "lat": 38.858,
      "lon": -77.085
    }
  ]
}
```

GET /arlington/pois/grocery-stores

Grocery store markers. Draw as green cart icons.

```
json
{
  "total": 208,
  "by_type": { "supermarket": 45, "convenience": 150, "greengrocer": 13 },
  "stores": [
    {
      "osm_id": 22222,
      "name": "Harris Teeter",
      "shop_type": "supermarket",
      "brand": "Harris Teeter",
      "address": "Pentagon City, Arlington, VA",
      "lat": 38.862,
      "lon": -77.059
    }
  ]
}
```

GET /arlington/base-layer

Lists all available layer endpoints. Good for health checking.

```
json
{
  "area": "Arlington, VA",
  "center": { "lat": 38.8816, "lon": -77.0910 },
  "cached_layers": ["boundary", "major_roads", "secondary_roads", "railways", "gas_s"]
}
```

GET /

Health check.

```
json
```

```
{
  "service": "Supply Chain Disruption API",
  "version": "0.1.0",
  "status": "running",
  "area": "Arlington, VA",
  "cached_layers": ["boundary", "major_roads", "secondary_roads", "railways", "gas_s
  "total_expected": 6
}
```

Fusion & Alert Endpoints

GET /arlington/alerts/active

Main endpoint for the alert panel. Runs fusion and returns all active disruption alerts. Each alert contains the 6 key fields.

json

```
{
  "total_alerts": 2,
  "alerts": [
    {
      "location": "I-395, Shirlington, Arlington, VA",
      "explanation": "NWS has issued a Flash Flood Warning for this area. Traffic ca
      "source_links": [
        {
          "source_name": "National Weather Service",
          "description": "Flash Flood Warning for Arlington County",
          "link": "https://api.weather.gov/alerts/active/zone/VAZ054"
        },
        {
          "source_name": "VDOT Camera arl_misc_4",
          "description": "flooding detected",
          "link": "C:\\...\\app\\data\\camera_images\\latest\\arl_misc_4.jpg"
        },
        {
          "source_name": "ARLnow",
          "description": "Flash Flooding Closes Multiple Roads in South Arlington",
          "link": "https://www.arlnow.com/flash-flooding-south-arlington"
        }
      ],
      "severity": 0.89,
      "confidence": 0.91,
      "recommended_actions": [
        "Reroute delivery trucks away from flooded road segments",
        "Alert gas stations and grocery stores of potential supply delays",
        "Coordinate with emergency services for road clearance timeline",
        "Identify alternate supply routes that bypass the flooded zone",
        "Priority: Monitor fuel availability at Shell, Exxon",
        "Priority: Monitor supply levels at Harris Teeter, Giant Food"
      ],
      "alert_tier": "CRITICAL",
      "affected_pois": [
        {
          "type": "gas_station",
          "name": "Shell",
          "address": "123 Route 1, Arlington, VA",
          "lat": 38.850,
          "lon": -77.051
        },
        {
```

```
    "type": "supermarket",
    "name": "Harris Teeter",
    "address": "Pentagon City, Arlington, VA",
    "lat": 38.862,
    "lon": -77.059
  }
],
"affected_roads": [
  { "name": "Henry G. Shirley Memorial Highway", "type": "motorway", "ref": "I"},
  { "name": "Richmond Highway", "type": "trunk", "ref": "US 1" }
],
"timestamp": "2026-04-01T05:30:00+00:00"
}
]
```

The 6 key fields per alert:

Field	Type	Description
location	string	Human-readable location (road names, area)
explanation	string	What's happening, evidence from each source, impact
source_links	array	Each source with source_name, description, and link
severity	float	0.0 (none) to 1.0 (catastrophic)
confidence	float	0.0 (guess) to 1.0 (confirmed)
recommended_actions	array	Specific steps to improve the situation

Alert tiers:

Tier	Color	Meaning
CRITICAL	Red  #f44336	High severity + multiple sources confirm
WARNING	Orange  #ff9800	Moderate severity + 2+ sources
WATCH	Yellow  #ffeb3b	Single source or low severity

GET /arlington/heatmap

For the map hexagon overlay. Returns deduplicated H3 cells with severity and boundary coordinates. Draw as colored hexagons on the map.

```
json
{
  "total_cells": 4,
  "cells": [
    {
      "h3_cell": "882a100d63ffffff",
      "tier": "CRITICAL",
      "severity": 0.89,
      "boundary": [
        { "lat": 38.8634, "lon": -77.0551 },
        { "lat": 38.8622, "lon": -77.0611 },
        { "lat": 38.8578, "lon": -77.0627 },
        { "lat": 38.8546, "lon": -77.0583 },
        { "lat": 38.8558, "lon": -77.0524 },
        { "lat": 38.8602, "lon": -77.0508 }
      ]
    }
  ]
}
```

Hexagon colors by severity:

Severity	Fill Color	Opacity
0.0 - 0.3	#ffeb3b yellow	0.2
0.3 - 0.6	#ff9800 orange	0.3
0.6 - 0.8	#f44336 red	0.4
0.8 - 1.0	#b71c1c dark red	0.5

Draw each cell as a polygon using the 6 boundary points.

GET /arlington/alerts/cell/{h3_cell}

Drill-down detail. When user clicks a hexagon on the map, call this with the h3_cell value.
Returns all raw signals and affected assets in that cell.

json

```
{
  "h3_cell": "882a100d63ffff",
  "signals": {
    "weather": [
      {
        "alert_type": "Flash Flood Warning",
        "severity": 0.95,
        "confidence": 1.0,
        "timestamp": "2026-04-01T04:00:00+00:00"
      }
    ],
    "traffic": [
      {
        "camera_id": "arl_misc_4",
        "congestion": "blocked",
        "anomaly": "flooding",
        "image": "C:\\...\\latest\\arl_misc_4.jpg",
        "severity": 0.90,
        "timestamp": "2026-04-01T04:10:00+00:00"
      }
    ],
    "news": [
      {
        "source": "ARLnow",
        "title": "Flash Flooding Closes Multiple Roads in South Arlington",
        "location": "Route 1, Pentagon City, Arlington VA",
        "event_type": "flooding",
        "severity": 0.85,
        "timestamp": "2026-04-01T04:20:00+00:00"
      }
    ]
  },
  "at_risk": {
    "pois": [
      {
        "type": "gas_station",
        "name": "Shell",
        "address": "123 Route 1, Arlington, VA",
        "lat": 38.850,
        "lon": -77.051
      }
    ],
    "roads": [
```

```
    { "name": "Richmond Highway", "type": "trunk", "ref": "US 1" }  
  ]  
}  

```

Frontend Loading Sequence

javascript

```

const API = "http://localhost:8000";

// 1. Load map config
const { center } = await fetch(`${API}/arlington/bbox`).then(r => r.json());

// 2. Initialize map
const map = L.map('map').setView([center.lat, center.lon], 13);

// 3. Load all base layers in parallel
const [boundary, majorRoads, secondaryRoads, railways, gasStations, groceryStores] =
  await Promise.all([
    fetch(`${API}/arlington/boundary`).then(r => r.json()),
    fetch(`${API}/arlington/roads/major`).then(r => r.json()),
    fetch(`${API}/arlington/roads/secondary`).then(r => r.json()),
    fetch(`${API}/arlington/railways`).then(r => r.json()),
    fetch(`${API}/arlington/pois/gas-stations`).then(r => r.json()),
    fetch(`${API}/arlington/pois/grocery-stores`).then(r => r.json()),
  ]);

// 4. Draw base layers + POI markers

// 5. Load heatmap overlay
const heatmap = await fetch(`${API}/arlington/heatmap`).then(r => r.json());
// Draw hexagons as colored polygons

// 6. Load alerts for side panel
const { alerts } = await fetch(`${API}/arlington/alerts/active`).then(r => r.json())
// Populate alert panel

// 7. On hexagon click → drill down
map.on('click', async (h3_cell) => {
  const detail = await fetch(`${API}/arlington/alerts/cell/${h3_cell}`).then(r => r.json());
  // Show detail popup
});

```

Dark Map Tile (free, no API key)

https://{s}.basemaps.cartocdn.com/dark_all/{z}/{x}/{y}{r}.png
 Attribution: '© OpenStreetMap © CARTO'

Color Scheme

Base Layers

Layer	Color	Weight
Boundary	 #4fc3f7 cyan	2.5, dashed
Motorways	 #ff9800 orange	2.5
Trunks	 #ffc107 amber	2.0
Primary roads	 #a5d6a7 light green	1.2
Secondary roads	 #81c784 green	1.0
Railways	 #ab47bc purple	1.5, dashed

POI Markers

POI	Color
Gas station	 #ff9800 orange
Supermarket	 #4caf50 green
Convenience store	 #8bc34a light green

POI markers change color when affected by an alert

Status	Color
Normal	default color
WATCH	 #ffeb3b yellow
WARNING	 #ff9800 orange
CRITICAL	 #f44336 red